

KHAIRY ISLAM SHAHEEN

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EDUCATION

Egypt-Japan University of Science and Technology (E-JUST)

Oct 2023 -- Feb 2028

Bachelor of Science in Computer Science & Engineering | cGPA: 3.2

Relevant Coursework: Data Structures (C), Advanced Programming (OOP Python), Computer Networks (CSE 328/329), Signals & Systems, Probability & Statistics, Electronics Engineering, Manufacturing Processes, Project Management

EXPERIENCE

Shaheen for Import & Export

Aug 2022 -- Present

IT & Digital Operations Lead | Tanta, Egypt

- Architected and deployed a production Laravel 12 e-commerce platform (shaheentire.com) with 100+ routes, bilingual AR/EN support via Spatie Translatable, Paymob payment gateway (HMAC SHA-512 verification), Google OAuth, and Filament 4 admin panel -- serving 519 products across all 27 Egyptian governorates
- Built a zero-trust WhatsApp AI chatbot (FastAPI + Gemini 1.5 Flash) with a 3-layer security architecture: environment-level phone whitelist, hardcoded forbidden-keyword interceptor (pre-LLM), and stateful per-user conversation sessions -- enabling non-technical staff to query enterprise Drive files via natural language
- Designed and executed a 4-phase enterprise cloud migration of 1,343 files into an 8-folder master Google Drive architecture with Arabic text normalization middleware, idempotent Smart Sync diffing, and an isolated Security Quarantine for credentials and Postman collections
- Engineered a zero-cost federated identity system by decoupling cPanel IMAP/SMTP email hosting from Google IAM, achieving enterprise RBAC (5 permission groups) without Google Workspace licensing fees -- deployed as mobile PWA to bypass native mail client setup friction for non-technical staff
- Established RACI-based operational framework across 8 non-overlapping verticals for a 10+ person marketing division, using Gherkin-style (Given/When/Then) workflow handoff protocols to eliminate role overlap and ambiguous task ownership
- Managed domain infrastructure (Cloudflare DNS/SSL, cPanel, Bitwarden), SSH deploy keys, auto-deploy pipelines, and nightly drift-check monitoring between server and repository

Digital Egypt Pioneers Initiative (DEPI) -- Round 4

Jun 2026 -- Present

Cybersecurity Intern | Remote

- Simulating secure network architectures, routing protocols (OSPF, EIGRP), and defense topologies in Cisco Packet Tracer as part of an advanced cybersecurity training track

PC Building Store

Jun 2019 -- Aug 2022

Technical Assistant | Tanta, Egypt

- Assembled and configured 40+ custom PC builds tailored to gaming, creative, and business requirements
- Diagnosed complex hardware and software issues including thermal management, BIOS conflicts, and driver compatibility
- Advised customers on cost-effective component upgrades, driving repeat business

PROJECTS

Scentura ERP -- Enterprise Resource Planning System

Django 6, DRF 3.17, React 19, TypeScript, TanStack Router, Tailwind v4, Recharts, Electron 41, PostgreSQL

- Architected a 22-app Django backend for a perfume manufacturing business covering accounts, inventory, orders, manufacturing, forecasting, suppliers, purchases, returns, profitability, marketing, and AI analytics
- Implemented event-sourced inventory (InventoryTransaction with signed +/- deltas) for full audit trail, FIFO OrderBatchAllocation for batch tracking, and a manufacturing pipeline: BulkTank → TankTransaction → ProductionBatch → BatchMaterialConsumption
- Built a pure-Python ridge regression forecasting engine (no numpy/sklearn) with Gaussian elimination, feature engineering (lag-1, lag-season, roll3, trend, stockout, marketing), and explainable ML with feature importance
- Enforced 8-role RBAC (Owner/Admin/Operations/Inventory/Manufacturing/Sales/Marketing/Finance/Viewer) with JWT access tokens (15min) + HttpOnly refresh cookies (7d)

- Developed an alternative frontend prototype (Aeron) with Vite + React 19 + shadcn/ui + Framer Motion featuring 23 lazy-loaded route modules

Customer Ordering System

FastAPI, SQLAlchemy, SQLite/PostgreSQL, Docker, React, JWT, HuggingFace API

- Designed a feature-based vertically sliced backend with 7 routers (auth, cart, catalog, events, inventory, orders, payment, tickets) and 5 services
- Implemented order state machine (PENDING → CONFIRMED → PROCESSING → SHIPPED → DELIVERED → REFUNDED + CANCELLED), UUID payment stubs with idempotency keys, and stale order sweep every 5 minutes
- Integrated HuggingFace distilbert for automated support ticket priority detection (URGENT/NORMAL)
- Built PII redaction middleware stripping emails, phones, and credit cards from logs/responses; PromptInjectionGuard middleware blocking injection attempts before LLM touchpoints
- Containerized PostgreSQL 15 Alpine via Docker Compose with persistent volume mounts

MRL Eye State Classification -- Drowsiness Detection

PyTorch, DenseNet121, CUDA 12.6, OpenCV, NVIDIA RTX 3050

- Fine-tuned a DenseNet121 model (pretrained ImageNet) for binary classification (awake vs sleepy) on the MRL Eye Dataset -- 15 epochs, Adam optimizer, NVIDIA RTX 3050 4GB VRAM
- Built a real-time webcam inference pipeline with Haar Cascade eye detection, temporal smoothing to prevent classification flicker, async inference for non-blocking feed, FPS counter, and visual + audible drowsiness alerts
- Developed 21 supporting scripts: augmentation variants, retrain iterations (v2-v4), quantization, benchmarking, and camera tests

ErgoX-UI -- Gaze-Controlled Accessible Interface

SwiftUI, ARKit TrueDepth Camera, iPhone 16 Pro

- Designed a gaze-controlled mobile interface based on CHI'25 "Stretch Gaze Targets Out" research paper
- Implemented 9 SwiftUI views: Home, Messages (quick replies), Media (music player with 5 Arabic tracks), Dashboard, Settings, Calibration, Gaze Components, Distance Engine, and Design System
- Engineered adjustable dwell-time selection (500--1000ms, default 700ms) for accessibility tuning
- Built a 53KB HTML/JS prototype replicating the full design system for cross-platform demonstration

Bellissimo Coffee Shop -- Digital Atmosphere

React 19, Vite, Tailwind v3, Framer Motion, Lenis

- Created a "Liquid Glass" design system with glassmorphism (SVG noise overlay, backdrop-filter blur) and a Circadian Lighting Engine that auto-switches theme (Morning/Afternoon/Evening) based on time of day
- Built a multi-phase BaristaButton animation (tamping → extracting → poured) using Framer Motion keyframe variants
- Implemented 7 components: Navbar, Hero, About, MenuHighlights, CoffeeRoastRating, StoreStatus, Footer

Arduino Robotics Competition -- 3 Autonomous Subsystems

Arduino C++, RemoteXY, NewPing, DHT11, MQ Gas Sensor, LDR, HC-SR04

- Developed 3 independent robot subsystems: Bluetooth-controlled arena robot with environmental sensing (temperature/humidity/gas), LDR-based differential-drive light-seeker, and ultrasonic wall-following maze solver using left-hand rule traversal algorithm

ESP32-CAM -- AI Smart Stick for Visually Impaired

ESP32-CAM, Embedded C, Computer Vision

- Engineered an academic project for an AI-powered navigation aid with custom embedded dataset research and edge-device model optimization under memory/processing constraints

Hospital Management System

Python, Flet (GUI), SQLite

- Built a desktop application with OOP class hierarchy (Person → Patient/Doctor/Admin), appointment scheduling with conflict detection, billing with invoice generation, and role-based dashboards (patient, doctor, admin)
- 40+ commits across 4 development branches (main, oop_integration, models-feature, gui-feature)

SWE Perfume Marketplace

Express 5, MongoDB/Mongoose, JWT, Cloudinary, Jest/Supertest

- Developed a multi-vendor e-commerce API with user/vendor/admin roles, vendor application workflow with admin approval, Cloudinary image upload pipeline, and celebrate/Joi request validation

Computer Networks Interactive Study Guide

Python (build script), Vanilla JavaScript, HTML/CSS, Web Audio API

- Built a 4-mode offline-first SPA (Tutorial, Flash Cards, Professor's Focus, Practice Quiz) generated by a Python script injecting 1600+ lines of JS into an HTML template
- Engineered a Pomodoro timer with SVG ring visualization, tab-visibility drift compensation, localStorage persistence across refreshes, and Web Audio API chime synthesis (C5+E5 sine oscillators)

NMEA Serial Radar Simulator

Python, Serial Communication, NMEA 0183

- Generated NMEA 0183 sentences (GGA GPS fix + HDT heading) with configurable COM port/ baud rate and threaded interactive lat/lon/heading control

CERTIFICATIONS & TRAINING

Cybersecurity Internship -- DEPI Round 4

Jun 2026 -- Present

Digital Egypt Pioneers Initiative

Flutter Mobile App Development (72 Hours)

Jul 2025 -- Oct 2025

YAT Learning Center, Tanta

Suspension, Steering, and Wheel Alignment Training	2025
GB Academy (Ghabbour Auto), Cairo -- Theoretical & Practical	
Full Stack Web Development (141 Hours)	Oct 2021
YAT Learning Center, Tanta	
Mastering Python Programming Fundamentals (32 Hours)	Jul 2020
YAT Learning Center, Tanta	

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, PHP, C++, C, Dart, Swift, SQL, Arduino C++, LaTeX, MATLAB
Backend Frameworks	FastAPI (ASGI/uvicorn), Django 6 + DRF 3.17, Laravel 12 (Livewire 3, Filament 4), Express 5
Frontend	React 19, Tailwind CSS v3/v4, Framer Motion, SwiftUI, Flet, Flutter, HTML5/CSS3
Databases	PostgreSQL, SQLite, MongoDB (Mongoose), SQLAlchemy ORM, Eloquent ORM
AI/ML	PyTorch (DenseNet121, CUDA 12.6), HuggingFace Inference API, Google Gemini API, OpenCV (Haar Cascade)
DevOps & Infra	Docker Compose, Git, Cloudflare (DNS/SSL), cPanel, ngrok, SSH deploy keys, SMB/NTFS ACLs
Automation	n8n workflow orchestration, Meta WhatsApp Cloud API, Google Drive API, shutil migration pipelines
Embedded Systems	ESP32-CAM, Arduino (RemoteXY, NewPing, DHT11, MQ, LDR, HC-SR04), NMEA 0183 serial protocol
Networking	Cisco Packet Tracer (OSPF, EIGRP), Nmap/Zenmap, Windows SMB infrastructure
Design	Figma, Adobe Photoshop, Adobe Illustrator, Canva, Pixlr
Productivity	Obsidian, Notion, Akiflow, Todoist, Overleaf (LaTeX)

LANGUAGES

Arabic	Native	English	Fluent	German	A2	French	A1--A2	Japanese	JLPT N5
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